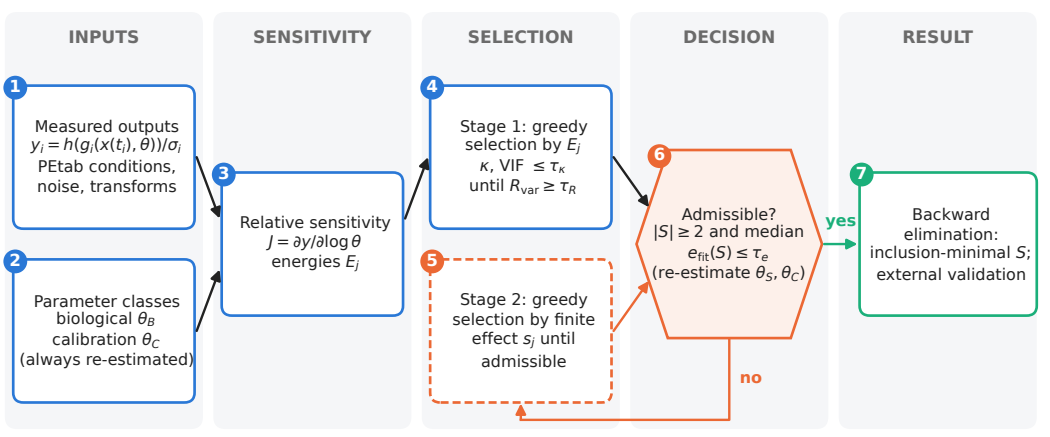
INPUTS
SENSITIVITY
SELECTION
DECISION
RESULT
1
Measured outputs
$y_i = h(g_i(x(t_i), \theta))/\sigma_i$
PEtab conditions,
noise, transforms
2
Parameter classes
biological θ_B
calibration θ_C
(always re-estimated)
3
Relative sensitivity
$J = \partial y/\partial \log \theta$
energies E_j
4
Stage 1: greedy
selection by E_j
κ, VIF $\leq \tau_\kappa$
until $R_{\text{var}} \geq \tau_R$
5
Stage 2: greedy
selection by finite
effect s_j until
admissible
6
Admissible?
$|S| \geq 2$ and median
$e_{\text{fit}}(S) \leq \tau_e$
(re-estimate θ_S, θ_C)
yes
no
7
Backward
elimination:
inclusion-minimal S;
external validation